

Advanced Public Economics — Applied Tutorial

Session IV: Inverse Optimum Approach

Daniel Weishaar

Roadmap for Session IV

Estimate the **shape of welfare weights** implied by status quo tax schedule.

- ① Invert optimal tax rate formula to get cumulative weights.
- ② Transform cumulative weights to individual weights.

From Optimal Taxes to Welfare Weights

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- ▶ estimated **income distributions, elasticities, hazard rates**, and
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Remember that under quasi-linear utility function, optimal tax rates under general welfare weights are described as

$$\frac{T'_{opt}(y)}{1 - T'_{opt}(y)} = (1 - \mathcal{G}(y)) \frac{1 - F_y(y)}{f_y(y)} \frac{\omega'_0(y)}{\omega_0(y)} \frac{1}{\varepsilon}$$

where $\mathcal{G}(y) = \int_y^\infty g(w_0(y')) f_y(y') dy'$ — the population-weighted welfare weight of people earning more than y .

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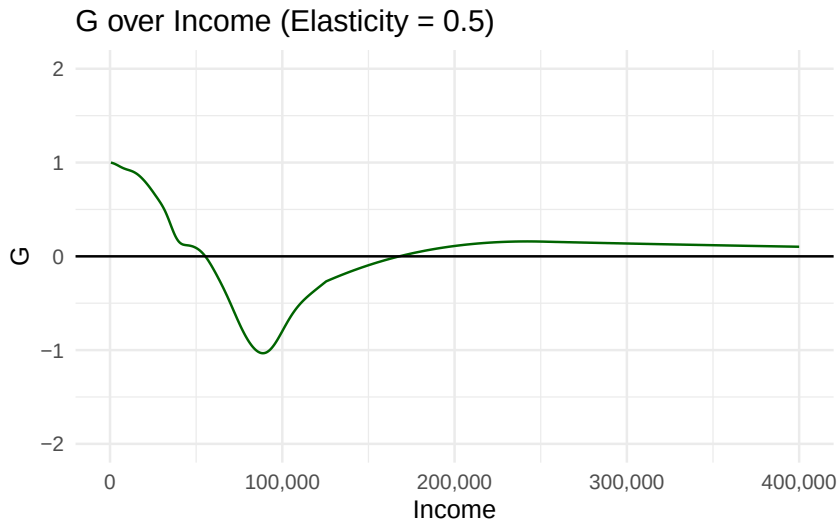
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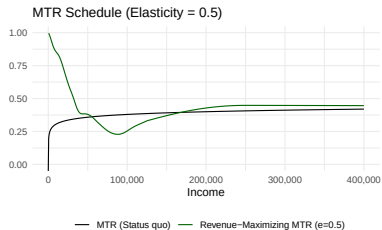
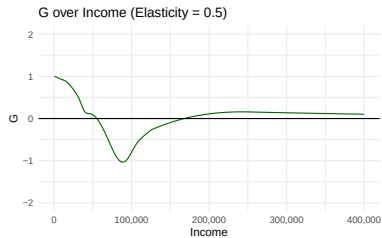
⇒ Hands-on exercise

Invert the Optimal Tax Rate Formula — $\mathcal{G}(y)$

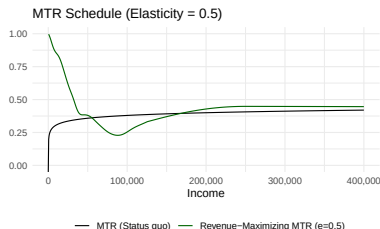
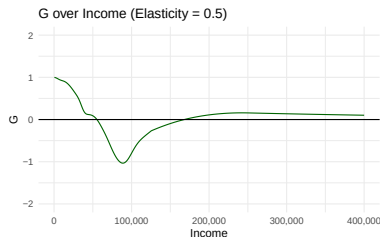


Using $\mathcal{G}(y)$ to identify inefficiently high MTR

Compare $\mathcal{G}(y)$ and the MTR schedule.



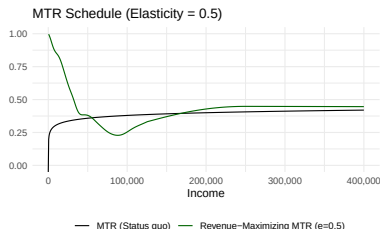
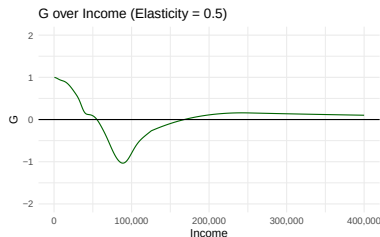
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- **Observation:** Whenever $\mathcal{G}(y) < 0$, the status quo MTR at income y exceeds the revenue maximizing rate. Thus, a small MTR cut at y would *increase* tax revenue.

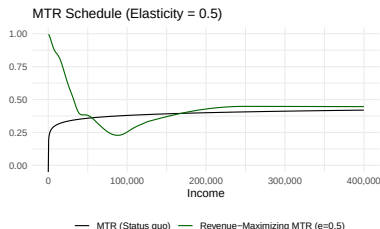
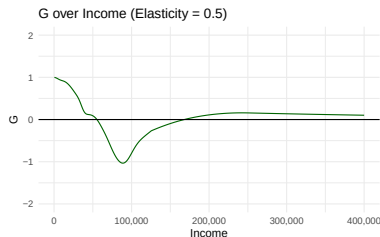
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- **Intuition:** To rationalize an observed tax schedule that lies above the revenue maximizing rate, the social planner “hates” to increase the consumption of people earning above 50,000 so much, such that she is even willing to sacrifice tax revenue to avoid that.

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⇒ **Question:** Which people are the ones with negative marginal weights?

Recovering Marginal Welfare Weights (I)

We are interested in the exogenous marginal welfare weight $g(w_0(y))$.

- ① Starting with definition of the cumulative welfare weight above income y as

$$\mathcal{G}(y) = \int_y^{\infty} g(w_0(t)) f_y(t) dt.$$

- ② Using the Leibniz rule,

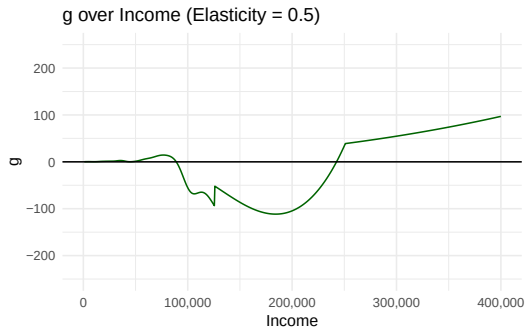
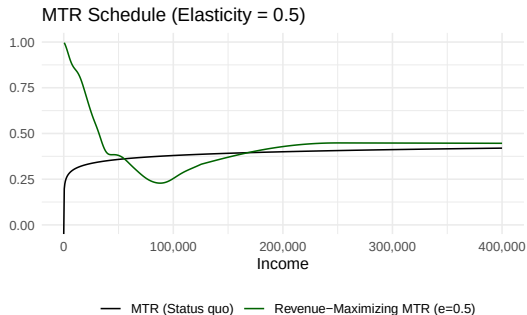
$$\frac{d}{dy} \mathcal{G}(y) = \frac{d}{dy} \int_y^{\infty} g(w_0(t)) f_y(t) dt = -g(w_0(y)) f_y(y).$$

- ③ Then, solve for the welfare weight

$$g(w_0(y)) = -\frac{\mathcal{G}'(y)}{f_y(y)} \quad \text{for } f_y(y) > 0$$

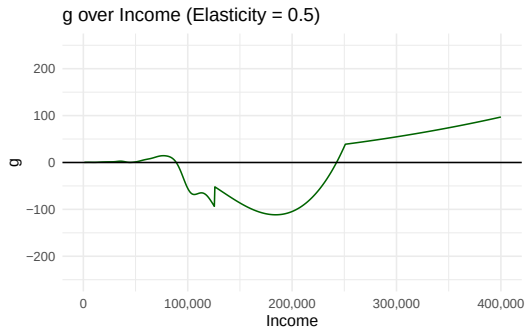
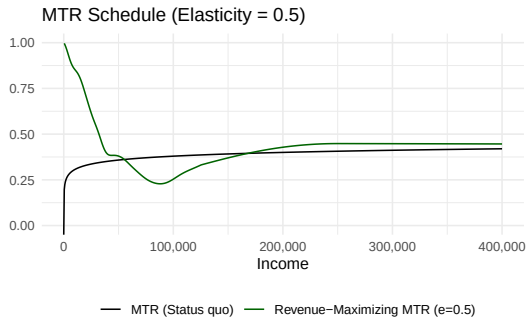
Note: $\mathcal{G}'(y)$ can be estimated using numerical differentiation, i.e., $\mathcal{G}'(y) \approx \frac{\Delta \mathcal{G}}{\Delta y}$.

Recovering Marginal Welfare Weights (II)



Question: Why is the implied marginal welfare weight for people with no income so small even though MTR in the status quo is so low?

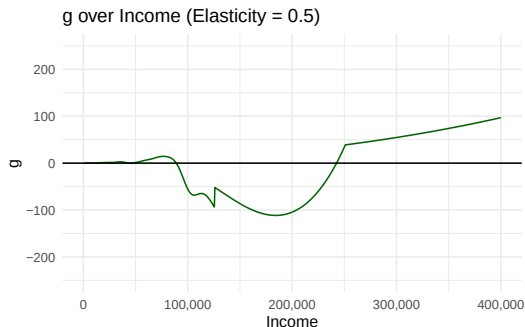
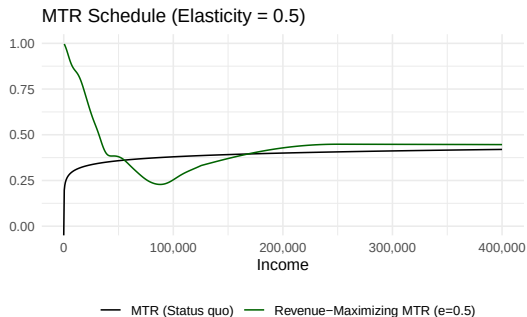
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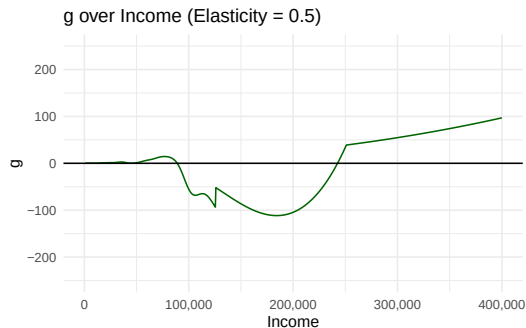
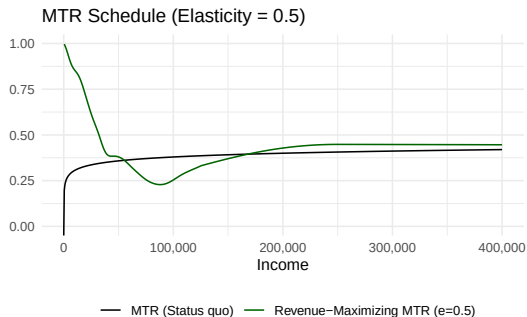
⇒ For those with $y = 0$, only lump-sum revenue (demogrant) matters. To maximize the demogrant, social planner uses very high MTR at the bottom, because they yield large revenue gains (since everyone with $y > 0$ must pay higher taxes then).

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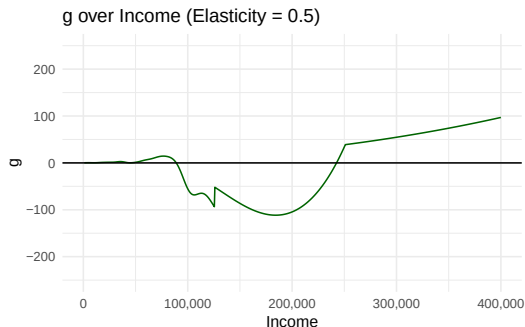
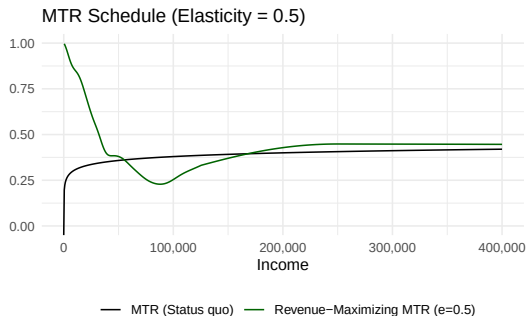
- ⇒ For those with $y = 0$, only lump-sum revenue (demogrant) matters. To maximize the demogrant, social planner uses very high MTR at the bottom, because they yield large revenue gains (since everyone with $y > 0$ must pay higher taxes then).
- ⇒ But, remember that statutory MTR used for this exercise does not capture full tax and transfer system (e.g. high transfer phaseout rates missing).

Recovering Marginal Welfare Weights (II)



Question: Why is the implicit welfare weight here positive for people at the top?

Recovering Marginal Welfare Weights (II)

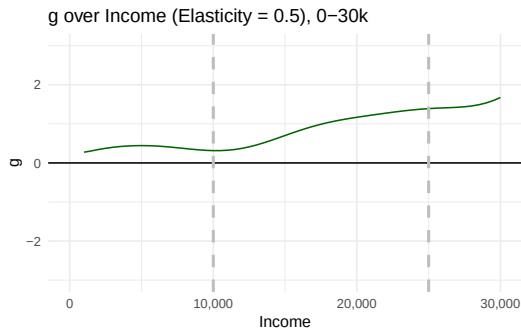


Question: Why is the implicit welfare weight here positive for people at the top?

⇒ The planner could extract more revenue by taxing people at the top more heavily. Since the planner does not do so in the status quo, he must put some positive value on the consumption of people at the top.

Interpreting Marginal Welfare Weights

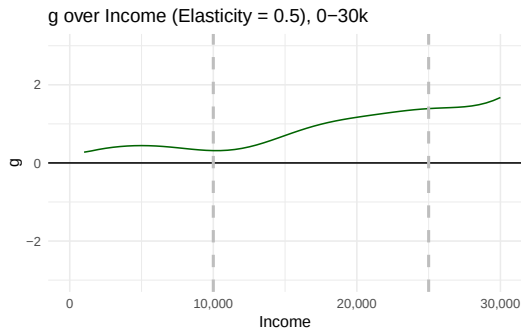
For instance, $g(10,000) = 0.32$ and $g(25,000) = 1.39$.



- **Interpretation 1:** Giving 1 EUR to person with $y = 10,000$ ($y = 25,000$) increases welfare by 0.32 (1.39) units.

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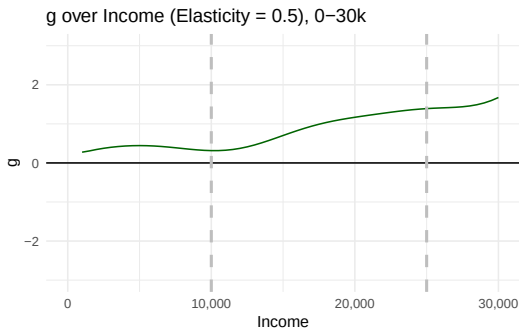
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- **Interpretation 3** (since marginal value of public funds = 1): To give someone with $y = 25,000$ 1 EUR, social planner is willing to sacrifice 1.39 EUR in public funds at the margin.

Some Words of Caution

- ▶ In my opinion, the inverse optimum approach is an illustrative exercise that can help us understand the redistributive preferences embedded in the tax code.
- ▶ Setting taxes is a complex process where multiple parties with different interests are involved (e.g., think about coalition governments, political economy constraints, etc.). **Recovered weights should not be taken literally.**
- ▶ Some research uses the inverse optimum approach to estimate implied **welfare weights by political parties**, e.g., by using the proposed tax rates from party manifestos before an election (Jacobs et al., 2017).
- ▶ Alternatively, recent research estimates welfare weights of the public more directly using **survey experiments** (Capozza and Srinivasan, 2025).

That's it! See you next time!

References

References I

-  BOURGUIGNON, F. and A. SPADARO (2012). “Tax–benefit revealed social preferences.” *The Journal of Economic Inequality* 10 (1), pp. 75–108.
-  CAPOZZA, F. and K. SRINIVASAN (2025). “Who should get money? Estimating welfare weights in the U.S.” Working Paper.
-  JACOBS, B., E. L. JONGEN, and F. T. ZOUTMAN (2017). “Revealed social preferences of Dutch political parties.” *Journal of Public Economics* 156, pp. 81–100.